Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element H Initial Template

How to use this template

Guidance before filling out the table on the next page:

- Enter *all* of the high-priority design requirements you listed in Element C into the table below.
- Determine an appropriate testing method for each requirement. For example, the test may involve measurement, inspection, physical actions, or mathematical modeling.
- As you plan, use your scientific method best practices, e.g. creating experimental controls, defining independent and dependent variables, only changing one variable at a time and changing it in increments to find a trend, state your prediction, etc.
- Your testing plan for a given requirement must include enough detail so that a person who is not on the design team could fully replicate the test. Include photos and/or diagrams to add clarity. Each test should include the following details:
    - how the test will be set up
    - what needs to happen during the test
    - test cases (if applicable)
    - number of trials of the test expected (aim for 7 or more trials)
    - the expected range of results that could be recorded, including units
    - the criteria for passing or failing the associated design requirement, including any permissible range for error and/or minimal thresholds
    - any goals your team has set beyond the passing result

Element H: Prototype Testing and Data Collection Plan

Testing Details *(H1, H2, H3, H4)*

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	Must float	1) the ROV must be able to float and sink when wanted. 2) we tested at the JCC and in the ocean. 3) we also used a bin filled with water.	a. 3 tests at the JCC and one in the ocean and once in a bin filled with water	Float Sink	**Fail**: sinks **Pass**: floats and stays where we want it to. **Goal**: for the ROV to be able to float and sink when we want/ us control its depth.
2	Collection system	1) The final trial in the ocean was how we tested the collection system of microplastics.	One test in the ocean.	Collect plastics in the ocean	**Fail**: collects no plastic **Pass**: collects plastics **Goal**: to collect microplastics in the ocean.
3					
4					

Element H continues on the next page →